

Ashwin K. Aggarwal

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EDUCATION

Cornell University – College of Engineering

Ithaca, NY

Bachelor of Science in Computer Science and Minor in Artificial Intelligence (GPA: 3.67)

Expected: May 2028

Relevant Coursework: Algorithms, Large-Scale Machine Learning, Deep Learning, Linear Algebra, Data Structures (Java), Probability and Stats, Computer Vision, Robotics, Computer Systems, Discrete Math, Functional Programming (OCaml)

EXPERIENCE

Virginia Commonwealth University

July 2026 – Aug 2026

Machine Learning Research Assistant, Wargo Lab

Richmond, VA

- Extended an FDA-developed simulation pipeline from 2D to 3D volumetric imaging using CUDA on an HPC cluster, generating 500+ synthetic scans with inserted anomalies and debugging a critical object-generation error
- Built concurrent ETL workflows to ingest network-device data into SQL Server and automate Power BI dashboards, replacing manual collection across 5+ client sites
- Co-authored a manuscript documenting the system design, automation workflow, and architecture for publication

Virginia Commonwealth University

May 2026 – Jun 2026

AI Engineering Intern, AI Department

Remote

- Designed a Databricks medallion pipeline for large-scale clinical text processing, structuring unstructured NoSQL data for downstream RAG and medical-query applications
- Used OCR models to extract 100+ fields from clinical documents for agentic diagnosis and demographic classification; reduced projected LLM API usage 90% through structured retrieval and prompt optimization

Cornell Computational Biology Lab

Jan 2026 – May 2026

Undergraduate Machine Learning Researcher

Ithaca, NY

- Built and fine-tuned 3D CNNs using ResNet and U-Net architectures to classify 500+ MRI scans; improved F1 from 0.54 to 0.61 through data augmentation, dropout, and early stopping in PyTorch
- Built preprocessing pipelines to normalize structural brain scans and evaluated 10+ model architectures to identify predictive signal

Next Play

Jul 2025 – Aug 2025

Software Engineering Intern, AI/ML Team

Remote

- Built a real-time Computer Vision pipeline in TypeScript and React using MediaPipe pose detection, supporting rep counting and live feedback across 20+ fitness challenges, achieving 90% pose classification accuracy for users
- Developed and shipped 10+ backend features and REST API integrations for a sports game in Golang and Kotlin, deployed on AWS, contributing to gameplay functionality and cross-service communication at a fast-moving startup

PROJECTS

Multi-Agent Research Assistant | MCP, LangGraph, Docker, Claude API, Neo4j, Python

- Built a four-agent orchestration system on the Claude API with LangGraph that ingests, summarizes, and cross-links 30+ research papers into a Neo4j knowledge graph, surfacing 100+ cross-document connections
- Automated nightly paper ingestion with APScheduler and an arXiv MCP server; shipped a Streamlit interface with interactive pyvis graph visualization for tracing citation and concept paths across the corpus

AI Housing Lease Analyzer | LangChain, FAISS, BM25, NLP, Python, OpenAI API, Docker

- Built a hybrid RAG pipeline combining FAISS vector search and BM25 retrieval with LangChain and OpenAI to answer questions over long, multi-page contracts, reaching 94% faithfulness on RAGAS evaluation
- Launched a Streamlit app enabling natural language contract queries, cutting manual lease review workload by 80%

AI Basketball Shot Tracker | Python (OpenCV, PyTorch, Numpy, Matplotlib), Ultralytics (YOLO), RoboFlow

- Fine-tuned YOLOv8 on 300+ hand-labeled frames for ball and hoop detection, achieving 98.5% and 98.0% mAP@50, respectively, using Roboflow for annotation and augmentation
- Built OpenCV pipeline for live ball and hoop tracking, achieving 90%+ accuracy in make/miss shot classification

SKILLS/INTERESTS

Languages: Python, C++, Java, C, C#, Go, TypeScript, Rust, OCaml

ML/AI: PyTorch, CUDA, OpenCV, YOLO, MCP, LangGraph, LangChain, RAG, FAISS, BM25, Neo4j

Infrastructure: Docker, Kubernetes, AWS, CI/CD, ROS, PostgreSQL, GCP

Interests: Catan, Physical AI, Philosophy, Basketball, Chess, Startups